

BRYAN PETRIE

SOFTWARE ENGINEERING & INFORMATION TECHNOLOGY

My Website: Bryan Petrie - Portfolio

CONTACT

319-930-6072

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SKILLS

- Python & Java
- SQL & Database Design
- Web Development (HTML/CSS)
- System Administration
- PowerShell Scripting
- VM/Hypervisor Management

EDUCATION

William Penn University

2024-2027

-BS in Software Engineering and Information Technology

-Minor in Business Management

(3.9 GPA)

-Dual sport athlete and a captain in soccer and track while completing studies

PROFILE

Software Engineering student with hands-on experience in full-stack development, system administration, and database management. Proven ability to develop user-friendly applications and solve complex technical problems. Strong communicator with experience training others and managing technical projects from concept to deployment.

EXPERIENCE

Line Cook at Buff's Gameday

Sept. 2022- Jan. 2026

Gained experience in a work establishment and developed skills in communication and work area management. Assisted in training new employees and establishing kitchen standards while maintaining health standards.

William Penn University Lab Assistant

2025-2026

Managed and assisted over 300 students with software and programming issues by guiding them through the processes so they understood how to navigate the problems themselves.

PROJECTS

AI CHATBOT DEVELOPMENT

Developed, trained using LLM and large-scale databases, and implemented chatbots for different local businesses to aid in customer support and employee onboarding.

POS DEVELOPMENT

Assisted in the implementation and development of a "Point of Service" system in a local establishment. Consulted staff to ensure the system was user friendly and flexible.

TASK MANAGER DEVELOPMENT

Programmed and deployed a personalized task manager onto PC. Used to not only manage tasks but increase productivity and performance of the user and machine. Also used to help manage multiple VM's at a time.

TECHNICAL SKILLS

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PROGRAMMING LANGUAGES
Python: Data processing, automation scripts, and application development
Java: Object-oriented design, software development, and debugging
SQL: Database queries, optimization, and relational database design
HTML/CSS: Responsive web design and user interface development
PowerShell: System automation and administrative task scripting

SYSTEMS & INFRASTRUCTURE
Virtual Machine Management: Creation, deployment, configuration, and maintenance of VMs
Network Administration: Configuration, troubleshooting, and security implementation
System Administration: Server management, user access control, and system monitoring
Database Management: Schema design, data modeling, query optimization, and backup procedures

DEVELOPMENT PRACTICES
Object-Oriented Programming: Design patterns, inheritance, polymorphism, and encapsulation
Version Control: Git for collaborative development and code management
Software Development Lifecycle: Requirements analysis, design, testing, and deployment
Agile Methodologies: Sprint planning, stand-ups, and iterative development

PROMPT ENGINEERING & AI INTEGRATION
Experienced in designing and optimizing prompts for large language models including GPT and Claude to maximize output quality and relevance. Proficient in implementing AI-assisted development workflows that enhance code quality, accelerate project delivery, and automate routine programming tasks. Skilled in integrating AI APIs into applications for natural language processing, automated content generation, and intelligent user interactions. Familiar with advanced prompting techniques including chain-of-thought reasoning, few-shot learning examples, and effective context management to achieve specific outcomes. Developed custom AI-powered tools for automated code documentation, debugging assistance, and unit test generation.

MICROSOFT AZURE CLOUD SERVICES
Hands-on experience with Azure Virtual Machine deployment, configuration, and management for building scalable cloud infrastructure solutions. Proficient in Azure Active Directory implementation for identity management, user authentication, and single sign-on configurations. Skilled in Azure SQL Database administration including schema design, backup strategies, performance tuning, and query optimization. Experienced in organizing and managing resource groups for efficient cloud resource allocation and cost management. Familiar with Azure DevOps for creating CI/CD pipelines that enable automated testing and deployment workflows. Knowledgeable in Virtual Network configuration to establish secure, cloud-based network architectures with proper subnet segmentation and security controls.